How-To Library

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Relationships and the Impact of Optimizing Relationships in TDM

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Abstract

This document explains how the relationships you assign between tables can impact the data in a subset operation in Test Data Management (TDM). It also explains with examples how the optimization feature helps reduce data returned in a subset operation. It assumes that you have knowledge of the data subset feature in TDM.

Supported Versions

- Test Data Management 10.2 .0 HotFix 2
- Test Data Management 10.2.0 HotFix 1
- Test Data Management 10.2 R1
- Test Data Management 10.2.0
- Test Data Management 10.1
- Test Data Management 9.7.1 HotFix 1
- Test Data Management 9.7.1
- Test Data Management 9.7.0
- Test Data Management 9.6.1 HotFix 1
- Test Data Management 9.6.1
- Test Data Management 9.6.1

Table of Contents

Overview. . . . 2
Major Edge. . . . 3
Minor Edge. . . . 4
Filter Propagation in an Entity Graph. . . . 5
Example 1. Diamond or Multipath Graph. . . . 5
Example 2. Self References. . . . 5
Example 3. Multi-table Cyclic. . . . 6
Extra Data in a Subset Operation. . . . 6
Wrong Edge Assignment. . . . 7
Correct Edge Assignment. . . . 8
Edge Optimization Options. . . . 8
Optimize Relations at the Entity or Plan Level. . . . 9
Data Integrity for Subset. . . . 9
Unexpected Data Subset Results. . . . 10
Conclusion. . . . 11

Overview

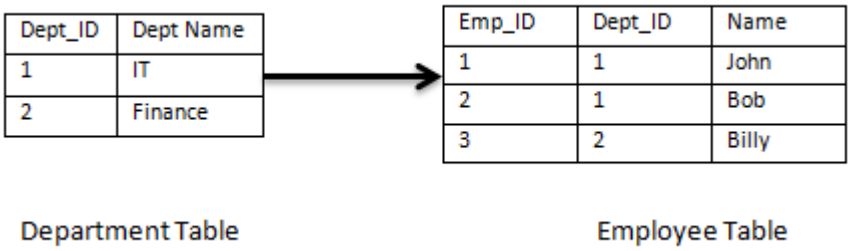
Tables in an entity might be related by physical or logical constraints. Constraints define parent-child relationships. The constraint might be a primary key constraint that is from the data source. The constraint might be a logical constraint. A logical constraint defines a parent-child relationship based on columns that are not keys in the data source.

Within the entity, all constraints are assigned a relationship severity, or edge behavior, which could be either major or minor. Edge is a TDM concept. The data selected in a subset could vary depending on the edge behavior assigned to each constraint. You can change the edge assigned to table constraints.

Major Edge

A major edge between two tables guarantees transactional integrity. Transactional integrity implies that in addition to selecting the parent records for a given record selected in the subset, all the child records of the parent records - with respect to the major edge - will also be selected.

Consider the following sample data:



Dept_ID	Dept Name
1	IT
2	Finance

Department Table

Emp_ID	Dept_ID	Name
1	1	John
2	1	Bob
3	2	Billy

Employee Table

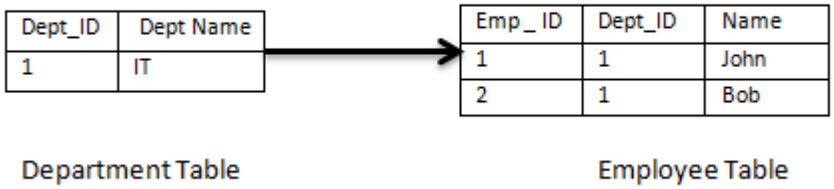
In the above figure, the arrow or edge represents a relationship between the Employee table and the Department table. Employee.Dept_ID is a foreign key that refers to Department.Dept_ID. In TDM, this edge could either be of major or minor type. Assume this edge is major. A major edge implies that when a department is selected, all its employees will also be selected.

Case 1. Apply the Criteria or Filter on the Child Table

```
Criteria = Employee.Name = 'Bob'
```

1. Bob is selected in the Employee table.
2. Department IT in the Department table is selected to ensure referential integrity.
3. Because the edge type is major, all child records must be selected. In this case, all employees with Dept Name IT are selected.

The final subset includes the following data:



Dept_ID	Dept Name
1	IT

Department Table

Emp _ ID	Dept_ID	Name
1	1	John
2	1	Bob

Employee Table

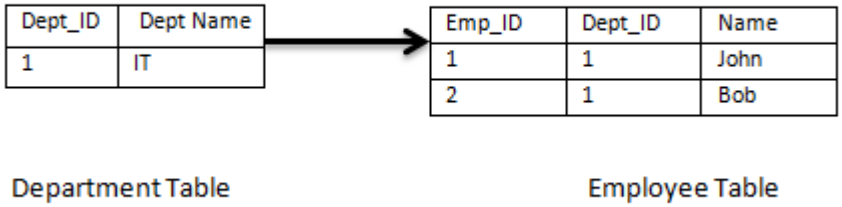
Note: Although the criteria was Name = 'Bob', to ensure transactional integrity, TDM includes all employees from Bob's department.

Case 2. Apply the Criteria or Filter on the Parent Table

```
Criteria = Department.Name = 'IT'
```

1. IT is selected in the Department table.
2. Bob and John are selected in the Employee table.

The final subset includes the following data:



Dept_ID	Dept Name
1	IT

Emp_ID	Dept_ID	Name
1	1	John
2	1	Bob

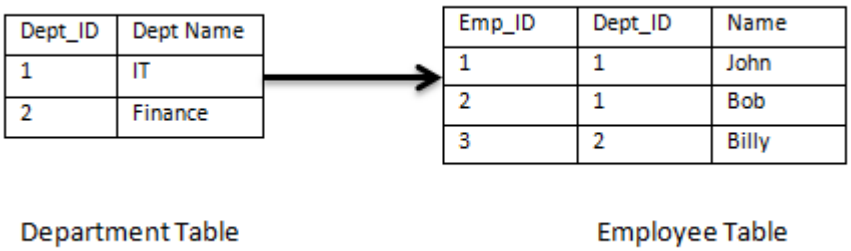
Department Table

Employee Table

Minor Edge

A minor edge between two tables guarantees only referential integrity. If a child record is selected in the subset, its parent record is also selected to maintain referential integrity. If the edge type is minor, selection in the parent table does not result in any selection in the child table.

Consider the data used in the previous example.



Dept_ID	Dept Name
1	IT
2	Finance

Emp_ID	Dept_ID	Name
1	1	John
2	1	Bob
3	2	Billy

Department Table

Employee Table

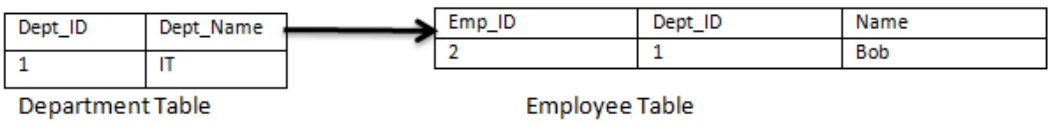
The arrow or edge represents a relationship between the Employee table and the Department table. Employee.Dept_ID is a foreign key that refers to Department.Dept_ID. Assume that this edge is minor.

Case 1. Apply the Criteria or Filter on the Child Table

```
Criteria = Employee.Name = 'Bob'
```

1. Bob is selected in the Employee table.
2. Department IT in the Department table is selected to ensure referential integrity.

The final subset includes the following data:



Dept_ID	Dept_Name
1	IT

Emp_ID	Dept_ID	Name
2	1	Bob

Department Table

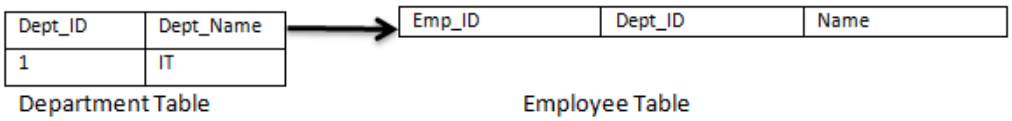
Employee Table

Case 2. Apply the Criteria or Filter on the Parent Table

```
Criteria = Department.Name = 'IT'
```

1. IT is selected in the Department table.

The final subset includes the following data:



Note: In this case, the result set does not include any data in the child table Employee.

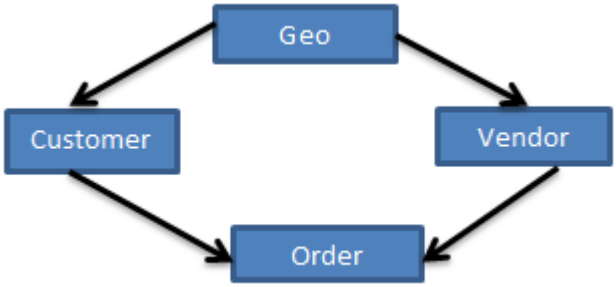
Filter Propagation in an Entity Graph

In an entity, the filter propagates from the table on which it is applied to various branches and includes all the nodes. Apart from the edge type, there are a number of factors that can influence data selection in a subset operation.

The following are a few common examples:

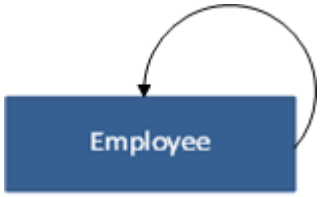
Example 1. Diamond or Multipath Graph

A diamond or multipath graph indicates a pattern when two or more paths exist to reach an object.



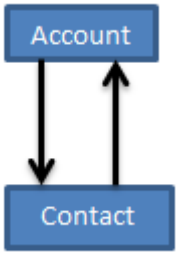
Example 2. Self References

An employee-manager relationship. If you select an employee in a subset, to ensure referential integrity, you must select the corresponding manager, and the manager's manager and so on. This leads to recursive selection of data. This results in more records getting selected in a table than expected based on the criteria.



Example 3. Multi-table Cyclic

Consider the following example. Every contact has an associated account and each account has a primary contact. Every time you select a contact, you must select the corresponding account, and then select the primary contact of the account and so on. This results in a cycle. Again, the number of records selected is more than you would expect based on the criteria.

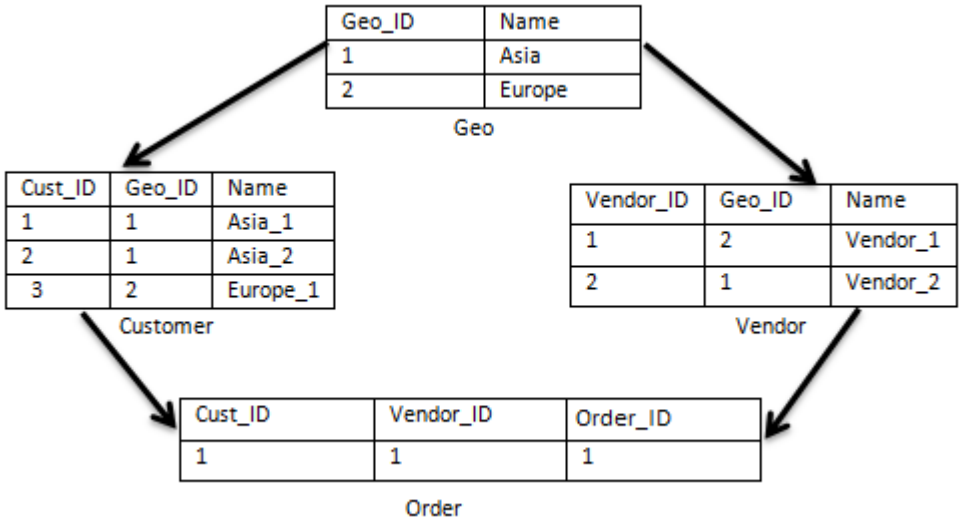


Extra Data in a Subset Operation

In some cases, the subset might include more data than expected. The selection of extra data can be because of the following two factors:

- The schema has a self-reference or a cycle. As explained in the previous section, this may lead to recursive selection of data and the subset may include more data than expected.
- The constraints have been assigned major edge behaviors in such a way that it leads to recursive data selection.

Consider the following example:



The Order table links customers and vendors. Both customers and vendors are related to the Geo table. Assuming criteria on the Geo table, at least two constraints must be assigned major edge behaviors in order to have subset selection for the Order table. If all the constraints are minor, the criteria will only affect the Geo table and will not propagate to Customer, Vendor or the Order table. There are two paths to reach the Order table from the Geo table.

Path 1 (Geo-Customer and Customer-Order Major)

If the Geo-Customer and Customer-Order constraints are assigned major behavior, the subset selection will proceed as follows:

All the customers for the geos selected by the criteria will be selected, and the orders placed by those customers will be selected because of the major behavior assignment. The vendors corresponding to the selected orders will also be selected to maintain referential integrity, and they will in turn select more geos. If the geos thus selected are the same as the ones selected initially by the criteria, the selection process stops here. If additional geos are selected in the last step, the selection process is triggered again due to the major edges and thus it becomes a recursive selection process and may lead to a larger subset than expected.

Assume `Criteria = Geo.Name = 'Asia.'`

1. Geo Asia is selected in the Geo table.
2. Customers with IDs 1 and 2 are selected in the Customer table.
3. The order placed by customer with name 'Asia_1' is selected.
4. The vendor for that order, that is vendor with ID =1 is selected.
5. Geo Europe is selected to maintain referential integrity.

The last step leads to selection of an extra Geo than is selected by the criteria to honor referential integrity. There is no way to avoid this selection. The extra geo triggers a recursive selection process which could select yet more data. There are ways to avoid this recursive selection. See the section on Edge Optimization Options in this document.

Path 2 (Geo-Vendor and Vendor-Order Major)

If the Geo-Vendor and Vendor-Order constraints are assigned major behavior, the subset selection will proceed as follows:

All the vendors for the Geos selected by the criteria are selected and the orders corresponding to those vendors are selected because of the major behavior assignment. The customers corresponding to the selected orders are also selected to maintain referential integrity and they will in turn select more geos. If the geos thus selected are the same as the ones selected initially by the criteria, the selection process stops here. If additional geos are selected in the last step, the selection process is triggered again due to the major edges and thus it becomes a recursive selection process.

Assume `Criteria = Geo.Name = 'Asia'`

1. Geo Asia is selected in the Geo table.
2. Vendor with ID 2 is selected in the Vendor table.
3. Selection ends here as vendor 2 has no associated orders. This path does not result in additional data.

Wrong Edge Assignment

A subset operation can result in additional data because of incorrect edge assignment.

Consider the following example:



Assume that we need to create a subset of all orders placed after 12/31/2013, customers who placed those orders, and the vendors who processed those orders.

The criteria in this case will be on the Order (child) table. For this example, both edges, or relationship types, must be minor. A minor edge guarantees referential integrity and will result in selecting only the customers, and vendors that are required.

If the Customer-Order edge is major.

All customers for selected orders are selected, and then all orders placed by those customers are selected. This second selection, which is required for transactional integrity guaranteed by the major definition, results in selection of unexpected orders, customers, and vendors.

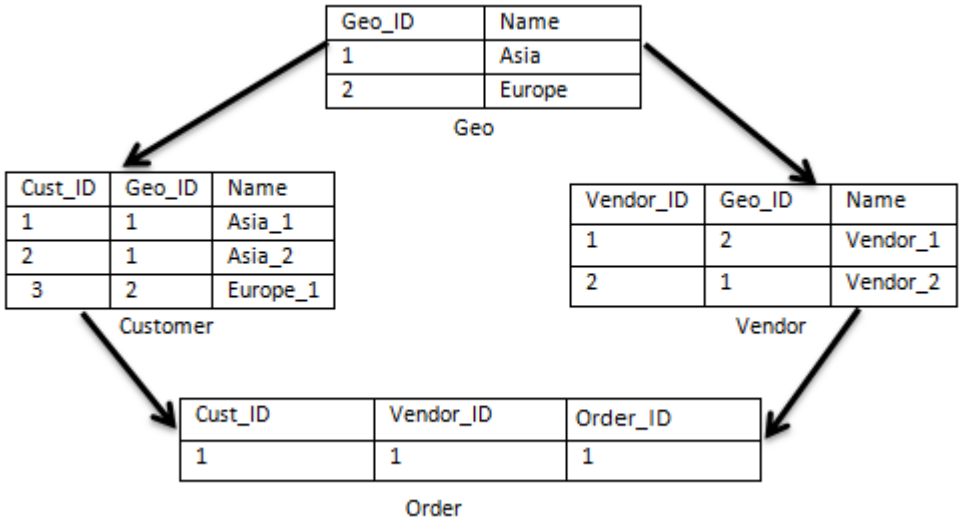
If the Order-Vendor edge is major.

Vendors for selected orders are selected. Because the Order-Vendor edge is major, all orders processed by the selected vendors are selected. This results in additional data in all three objects.

Correct Edge Assignment

A subset operation can also result in additional data when the edge assignment is correct but the definition of major edge results in additional data.

Consider the following sample data:



To trigger data selection, one path between the Geo and Order tables must have a major edge. Assume that the Geo-Customer-Order path is major and the Geo-Vendor-Order path is minor. This results in an extra geo Europe being selected. The Geo-Customer path is major. This mandates that all customers from Europe be selected. Again, this selection can result in additional data in all tables.

Edge Optimization Options

Two optimization options have been added in 9.6.1. These options can reduce data fetched in a subset operation.

Optimize Relations at the Entity or Plan Level

The Optimize Relations feature allows you to optimize relations either at the entity or at the plan level. When you optimize relations at the entity level, the changes are saved to the entity. When you optimize relations at the plan level, that is at run time, the changes impact only the entity in the operation.

When you use this feature at the:

- **Entity Level.** TDM optimizes relationships for the entity on which optimization is applied.
- **Plan Level.** TDM optimizes relationships for all entities in the plan.

When you enable this option, TDM traverses an entity starting from the criteria node. TDM attempts to convert maximum number of edges to minor while ensuring to include all the nodes in the graph. This results in the following improvements:

- Self references are converted to minor.
- In cases of diamond or multipath graphs, only one path remains major. This results in significant reduction in data.
- In cases that traverse up, that is child to parent, the edge types are converted to minor.

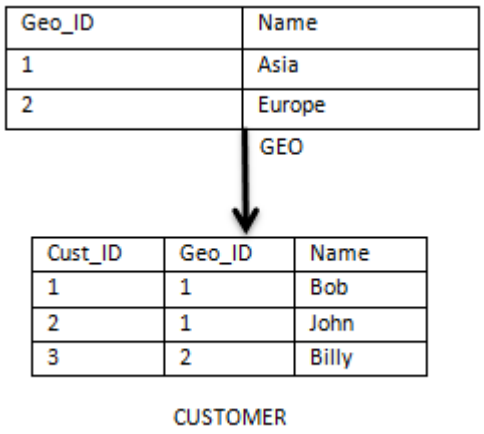
Data Integrity for Subset

Select the **Data Integrity for Subset** option in the **Advanced Settings** page of the plan settings.

The Data Integrity for Subset feature has the following options:

- **Transactional Integrity for Major type.** This is the default value. When you select this option, major edge behavior always conforms to definition. That is, for each parent record selected, all its child records are also selected.
- **Referential Integrity only.** This alters major edge behavior and treats it as once traversable only. Note that this will compute the subset through SQLs only and is not recommended for complex entities (typically more than 30-40 relationships). TDM ignores this option if the number of tables in an entity is larger than 25.

Consider the following example:



Configuration:

Data Integrity for Subset - referential Integrity only.

Relationship Type: Major

Criteria: On the child table `Customer.Name = 'Bob'`.

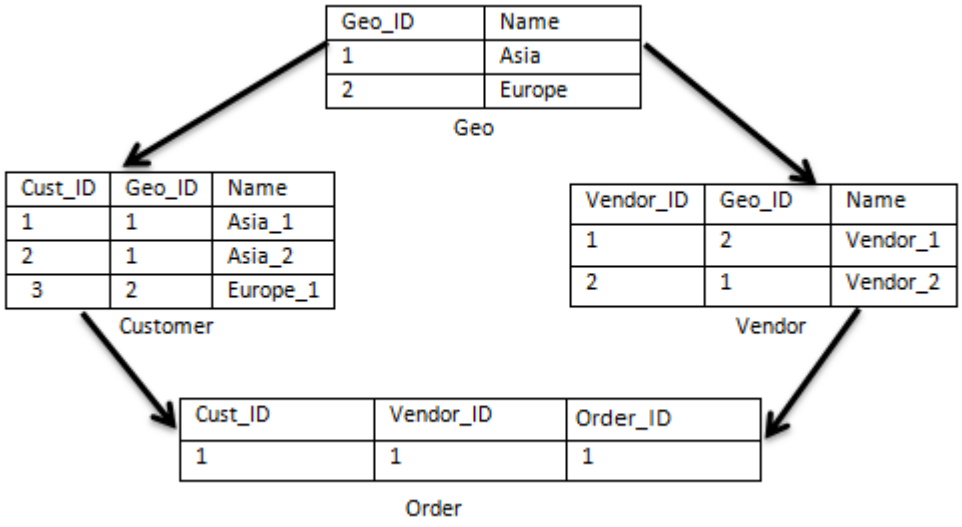
Selection:

1. Bob is selected in the Customer table.
2. Asia is selected in the Geo table.

Note: Although the relationship is major, the edge was traversed only once and all customers from the Geo were selected.

Example with criteria on parent table:

Consider the following data:



An additional Geo Europe is selected to maintain referential integrity. Typical major edge behavior would mean that customers for Europe are selected again. However, if you select the Referential integrity only option, selection stops at the Geo table as all major edges will now be once traversable only. Because selection has already happened for that edge once, no additional records are selected.

Unexpected Data Subset Results

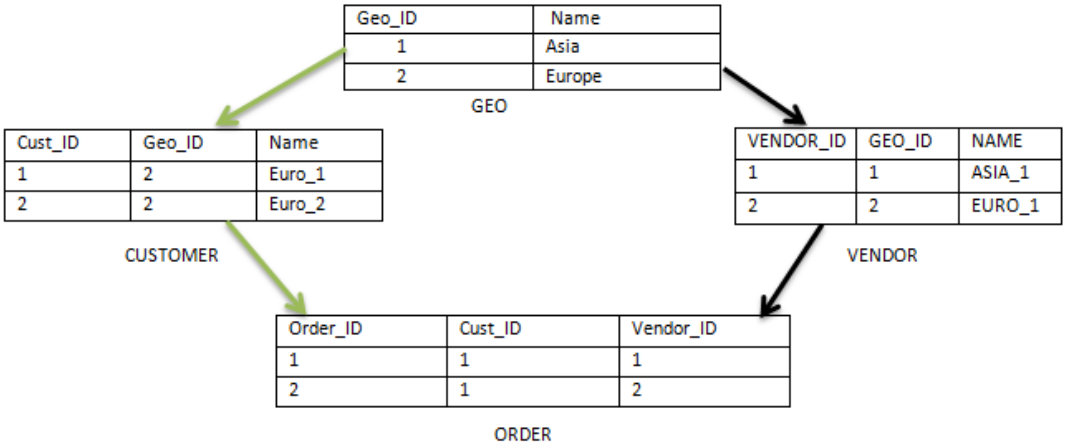
Unexpected results of a data subset operation can include both additional data being fetched and no data being fetched in the operation.

No Data

The optimization context selects paths and assigns major edge behaviors based on the structure of the schema and has no information about the data. In a diamond scenario, this may lead to selection of paths which have tables with no data. In such a case, the resulting subset may not have any data.

In the following example, the black edges represent minor relationships, and the green edges represent major relationships.

```
Criteria = Geo.Name = 'Asia'
```



There are two paths between the Geo and Order tables. Optimization makes one path major and the other path minor. Data selection starts from the major path and propagates to other nodes. The Geo-Customer-Order path is major, and the Geo-Vendor-Order path is minor.

Note that no customers exist for Geo = Asia. Therefore no selection happens in the Customer table. This results in no selection in the Order and Vendor tables. Although there are vendors for Geo = Asia, in this case, because selection happened from the other path, no data is fetched from any other table.

Additional Data

Assume that the Geo-Vendor-Order path is major while the Geo-Customer-Order path is minor.

Selection happens in the following order:

1. Geo = 'Asia' is selected.
2. Vendor_ID 1 for Asia_1 is selected in the Vendor table.
3. Order with ID = 1, related to vendor 'Asia_1' is selected.
4. Customer with Cust_ID = 1 is selected in the Customer table.
5. Customer 'Euro_1's Geo, Europe, is selected for referential integrity in the Geo table.
6. Because the Geo-Vendor edge is major, vendors for Europe are selected from the Vendor table. In this case, the vendor with ID = 2 is selected.
7. The order with ID = 2, related to Vendor.Vendor_ID = 2 is selected from the Order table.
8. The related customer is already selected. Therefore, the selection stops here.

Note: Even after optimization, an additional European vendor and the order related to the additional vendor are selected. These are not related to the initial criteria.

Conclusion

Optimizing relationships results in reduced data and faster execution. However, because optimization is purely metadata based, some selection can result in no data being selected. The Data Integrity for Subset with referential integrity option will result in a subset closer to the criteria. However, because it does all computations through complex SQL commands, it is not recommended for use with bigger entities.

Author

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